

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 May/June 2023 Worked Solutions

May/June 2023 paper rebuilt with the worked solution after each question.

10

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P1**

PAST PAPER

WMA11/01 May/June 2023

May/June 2023 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1**Inequalities****1.****In this question you must show all stages of your working.****Solutions relying on calculator technology are not acceptable.**

Solve the inequality

$$4x^2 - 3x + 7 \geq 4x + 9$$

(4)**(Total for Question 1 is 4 marks)**

Worked Solution - Question 1

1. Move all terms to one side

$$4x^2 - 3x + 7 \geq 4x + 9 \text{ gives } 4x^2 - 7x - 2 \geq 0.$$

2. Factorise

$$4x^2 - 7x - 2 = (4x + 1)(x - 2).$$

3. Use the sign of the quadratic

The roots are $x = -\frac{1}{4}$ and $x = 2$. Since the quadratic opens upwards, it is non-negative outside the roots.

Final answer

$$x \leq -\frac{1}{4} \text{ or } x \geq 2.$$

Question 2

Quadratics**2.****In this question you must show all stages of your working.****Solutions relying entirely on calculator technology are not acceptable.**

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350m,

(a) write down an equation linking x and y

(1)

Given also that the area of the pitch is 7350m^2

(b) write down a second equation linking x and y

(1)

(c) hence find the value of x and the value of y

(4)

(Total for Question 2 is 6 marks)

Worked Solution - Question 2

1. Use the perimeter

$$2x + 2y = 350, \text{ so } x + y = 175.$$

2. Use the area

$$xy = 7350.$$

3. Form a quadratic

$$\text{Using } y = 175 - x, x(175 - x) = 7350, \text{ so } x^2 - 175x + 7350 = 0.$$

4. Solve

$$x = \frac{175 \pm \sqrt{175^2 - 4(7350)}}{2} = \frac{175 \pm 35}{2}, \text{ giving } x = 105 \text{ or } 70.$$

5. Use $x > y$

$$\text{Since } x > y, x = 105 \text{ and } y = 70.$$

Final answer

$$x = 105, y = 70.$$

Question 3

Quadratics

3. (a) Express $3x^2 + 12x + 13$ in the form

$$a(x + b)^2 + c$$

where a , b and c are integers to be found.

(3)

- (b) Hence sketch the curve with equation $y = 3x^2 + 12x + 13$

On your sketch show clearly

- the coordinates of the y intercept
- the coordinates of the turning point of the curve

(3)

(Total for Question 3 is 6 marks)

Worked Solution - Question 3

1. Complete the square

$$3x^2 + 12x + 13 = 3(x^2 + 4x) + 13.$$

2. Use $(x + 2)^2$

$$x^2 + 4x = (x + 2)^2 - 4, \text{ so } 3(x^2 + 4x) + 13 = 3(x + 2)^2 - 12 + 13.$$

3. Read the features

$y = 3(x + 2)^2 + 1$ has turning point $(-2, 1)$. The y-intercept is found from $x = 0$, giving $y = 13$.

4. Sketch description

The curve is an upward-opening parabola with minimum $(-2, 1)$ and y-intercept $(0, 13)$.

Final answer

(a) $3(x + 2)^2 + 1$.

(b) y-intercept $(0, 13)$ and turning point $(-2, 1)$.

Question 4

Differentiation

4. In this question you must show all stages of your working.

(a) Write

$$y = \frac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}}$$

in the form

$$y = Ax^p + Bx^q$$

where A , B , p and q are constants to be found.

(4)

(b) Hence find $\frac{dy}{dx}$ giving each coefficient in simplest form.

(3)

(Total for Question 4 is 7 marks)

Worked Solution - Question 4

1. Rewrite the denominator

$$\sqrt[3]{8x} = 2x^{1/3}.$$

2. Rewrite the numerator

$$\sqrt{x^3} = x^{3/2}.$$

3. Divide each term

$$y = \frac{5x^2}{2x^{1/3}} + \frac{x^{3/2}}{2x^{1/3}} = \frac{5}{2}x^{5/3} + \frac{1}{2}x^{7/6}.$$

4. Differentiate

$$\frac{dy}{dx} = \frac{5}{2} \cdot \frac{5}{3}x^{2/3} + \frac{1}{2} \cdot \frac{7}{6}x^{1/6}.$$

5. Simplify coefficients

$$\frac{dy}{dx} = \frac{25}{6}x^{2/3} + \frac{7}{12}x^{1/6}.$$

Final answer

$$(a) y = \frac{5}{2}x^{5/3} + \frac{1}{2}x^{7/6}.$$

$$(b) \frac{dy}{dx} = \frac{25}{6}x^{2/3} + \frac{7}{12}x^{1/6}.$$

Question 5

Radians

5.

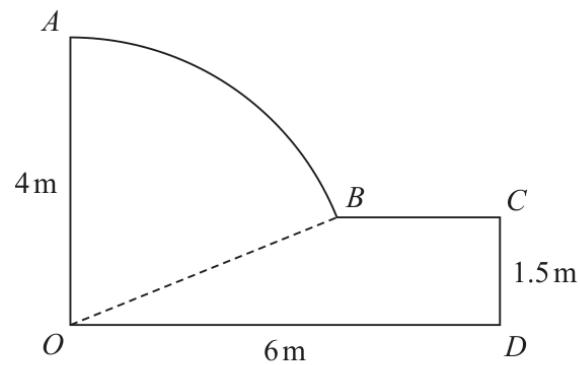


Figure 1

Figure 1 shows the plan for a garden.

In the plan

- OA and CD are perpendicular to OD
- AB is an arc of the circle with centre O and radius 4 metres
- BC is parallel to OD
- OD is 6 metres, OA is 4 metres and CD is 1.5 metres

(a) Show that angle AOB is 1.186 radians to 4 significant figures.

(2)

(b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.

(4)

(c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

(4)

(Total for Question 5 is 10 marks)

Worked Solution - Question 5

1. Find the angle AOB

Since $OB = 4$ and the height of B is 1.5 , $\cos \angle AOB = \frac{1.5}{4} = 0.375$.

2. Calculate the angle

$\angle AOB = \cos^{-1}(0.375) = 1.186 \dots$ rad, which is **1.186** rad to 4 significant figures.

3. Find BC

The horizontal coordinate of B is $4 \sin(1.186 \dots) = 3.708 \dots$, so
 $BC = 6 - 3.708 \dots = 2.2919 \dots$ m.

4. Find the arc length

Arc $AB = r\theta = 4(1.186 \dots) = 4.7456 \dots$ m.

5. Find the perimeter

Perimeter

$= OA + \text{arc } AB + BC + CD + OD = 4 + 4.7456 \dots + 2.2919 \dots + 1.5 + 6 = 18.5375 \dots$ m.

6. Find the area

Area of sector $AOB = \frac{1}{2}(4^2)(1.186 \dots) = 9.491 \dots$

7. Add the quadrilateral area

Area of $OBCD$ is the trapezium area $\frac{1}{2}(6 + 2.2919 \dots)(1.5) = 6.2189 \dots$

Total area $= 15.710 \dots = 15.7 \text{ m}^2$.

Final answer

(a) $\angle AOB = 1.186$ rad.

(b) 18.5 m.

(c) 15.7 m^2 .

Question 6

Indices & Surds

6.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

(b) Express $\frac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

(Total for Question 6 is 7 marks)

Worked Solution - Question 6

1. Expand the square

$$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2}.$$

2. Rationalise the denominator

$$\frac{1}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}} = \frac{3 - 2\sqrt{2}}{9 - 8} = 3 - 2\sqrt{2}.$$

3. Let r be the square root

Let $r = \sqrt{3 + 2\sqrt{2}}$. Then $r^2 = 3 + 2\sqrt{2}$ and $\frac{1}{r^2} = 3 - 2\sqrt{2}$.

4. Use part (a)

$$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2} = (3 + 2\sqrt{2}) - 2 + (3 - 2\sqrt{2}) = 4.$$

5. Take the positive square root

Since $r > 1$, $r - \frac{1}{r} > 0$, so $r - \frac{1}{r} = 2$.

Final answer

(a) $r^2 - 2 + \frac{1}{r^2}$.

(b) $3 - 2\sqrt{2}$.

(c) The expression equals 2.

Question 7

Inequalities

7.

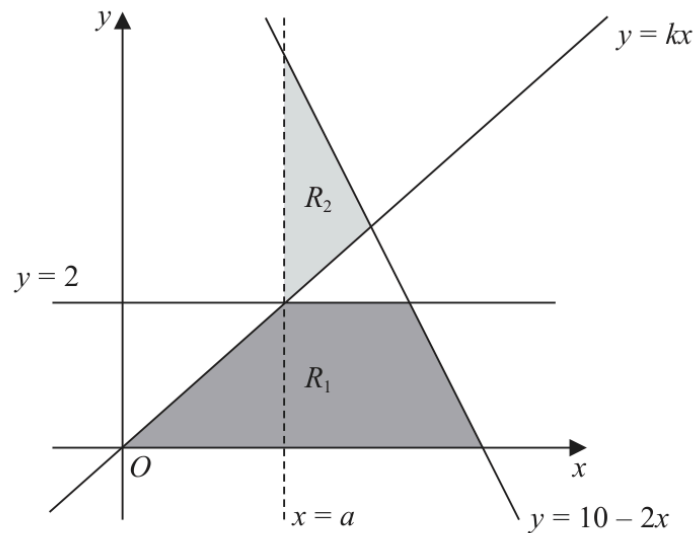


Figure 2

The region R_1 , shown shaded in Figure 2, is defined by the inequalities

$$0 \leq y \leq 2 \quad y \leq 10 - 2x \quad y \leq kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

(i) the value of a

(ii) the value of k

(4)

(b) Define the region R_2 , also shown shaded in Figure 2, using inequalities.

(2)

(Total for Question 7 is 6 marks)

Worked Solution - Question 7

1. Find the area of R_1

The bottom horizontal length is 5, from $x = 0$ to the intercept of $y = 10 - 2x$ at $x = 5$.

2. Use the top length

The top horizontal length is $4 - a$, because $y = 2$ meets $y = 10 - 2x$ at $x = 4$.

3. Solve for a

$$\text{Area } R_1 = \frac{1}{2}(5 + 4 - a)(2) = 9 - a = \frac{27}{4}, \text{ so } a = \frac{9}{4}.$$

4. Find k

The line $x = a$ passes through the intersection of $y = 2$ and $y = kx$, so $2 = ka$.

$$\text{Hence } k = \frac{2}{9/4} = \frac{8}{9}.$$

5. Define R_2

Region R_2 is to the right of $x = a$, above $y = kx$, and below $y = 10 - 2x$.

Final answer

$$(a) a = \frac{9}{4}, k = \frac{8}{9}.$$

$$(b) x \geq \frac{9}{4}, y \geq \frac{8}{9}x, y \leq 10 - 2x.$$

Question 8

Integration

8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

- (a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{\frac{1}{2}}$$

at the point $P(4, 12)$ Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{\frac{1}{2}}$$

- (b) find
- $f(x)$
- giving the coefficients in simplest form.

(5)

(Total for Question 8 is 10 marks)

Worked Solution - Question 8

1. Differentiate the curve

$$\text{For } y = \frac{1}{4}x^3 - 8x^{-1/2}, \frac{dy}{dx} = \frac{3}{4}x^2 + 4x^{-3/2}.$$

2. Find the gradient at P

$$\text{At } x = 4, \frac{dy}{dx} = \frac{3}{4}(16) + 4(4^{-3/2}) = 12 + \frac{1}{2} = \frac{25}{2}.$$

3. Find the tangent

$$\text{Using } P(4, 12), y - 12 = \frac{25}{2}(x - 4).$$

4. Write in integer form

$$\text{Multiplying by 2 gives } 2y - 24 = 25x - 100, \text{ hence } 25x - 2y - 76 = 0.$$

5. Integrate $f'(x)$

$$f'(x) = \frac{1}{4}x^3 - 8x^{-1/2} \text{ gives } f(x) = \frac{x^4}{16} - 16\sqrt{x} + c.$$

6. Use P(4, 12)

$$12 = \frac{4^4}{16} - 16\sqrt{4} + c = 16 - 32 + c, \text{ so } c = 28.$$

Final answer

$$(a) 25x - 2y - 76 = 0.$$

$$(b) f(x) = \frac{x^4}{16} - 16\sqrt{x} + 28.$$

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9 marks

Question 9

Trigonometric Functions

9. (i)

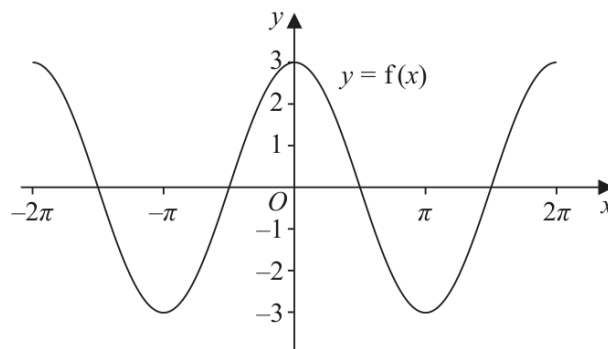


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

- (a) Write down an expression for $f(x)$ (2)

On a separate diagram,

- (b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

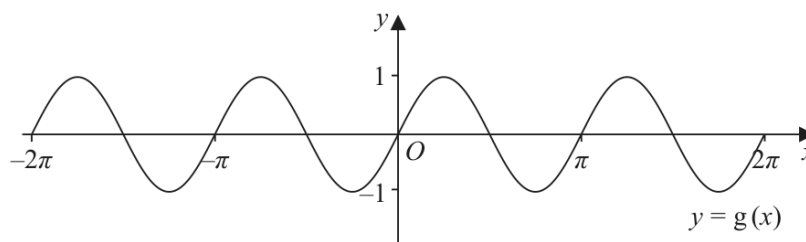


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

- (a) Write down an expression for $g(x)$ (2)

On a separate diagram,

- (b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(3)

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

1. Identify $f(x)$

The first graph has amplitude 3, period 2π , and maximum at $x = 0$, so $f(x) = 3 \cos x$.

2. Transform f

$$y = f\left(x + \frac{\pi}{4}\right) = 3 \cos\left(x + \frac{\pi}{4}\right), \text{ a shift left by } \frac{\pi}{4}.$$

3. Find the x-intercepts

$$\cos\left(x + \frac{\pi}{4}\right) = 0 \text{ gives } x = \frac{\pi}{4} + n\pi. \text{ In } [-2\pi, 2\pi]$$

4. Find the y-intercept

$$\text{At } x = 0, y = 3 \cos \frac{\pi}{4} = \frac{3\sqrt{2}}{2}.$$

5. Identify $g(x)$

The second graph has amplitude 1, period π , and passes through the origin with positive gradient, so $g(x) = \sin 2x$.

6. Transform g

$y = g(x) - 2 = \sin 2x - 2$, so the graph is shifted down by 2. Its y-intercept is $(0, -2)$.

Final answer

(i)(a) $f(x) = 3 \cos x$. (i)(b) x-intercepts $-\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$ and y-intercept $\frac{3\sqrt{2}}{2}$. (ii)(a) $g(x) = \sin 2x$. (ii)(b) y-intercept $(0, -2)$.

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10 marks

Question 10

Straight Line

10.

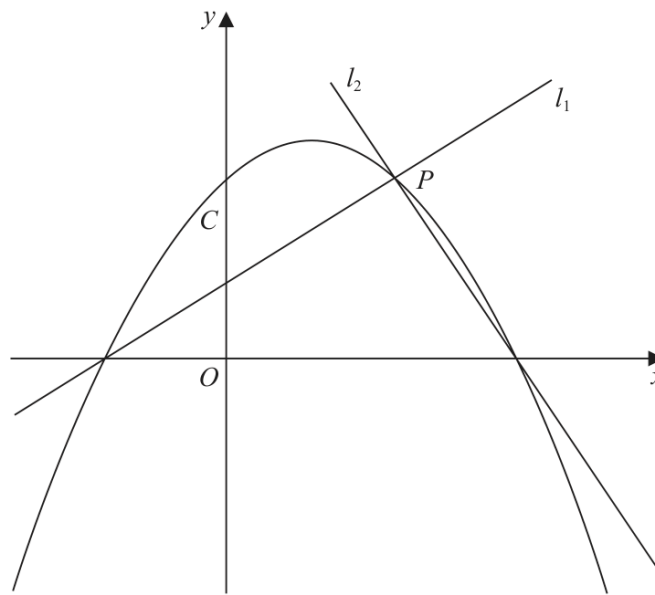


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x -axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b , the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

(b) find, in terms of b , an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Find l_1

Line l_1 has gradient $\frac{1}{2}$ and passes through the negative x-axis root $(-2, 0)$, so

$$y = \frac{1}{2}(x + 2) = \frac{1}{2}x + 1.$$

2. Find P

Set $-\frac{1}{4}(x + 2)(x - b) = \frac{1}{2}(x + 2).$

3. Solve the intersection

Multiplying by -4 gives $(x + 2)(x - b) = -2(x + 2)$, so
 $(x + 2)(x - b + 2) = 0.$

4. Choose the second point

The root $x = -2$ is the x-axis point, so P has $x = b - 2$. Then

$$y = \frac{1}{2}(b - 2) + 1 = \frac{b}{2}.$$

5. Find l_2 from the positive root

Since l_2 is perpendicular to l_1 , its gradient is -2 . It passes through the positive x-axis root $(b, 0)$, so $y = -2(x - b) = -2x + 2b$.

6. Use P to write another l_2

Through $P\left(b - 2, \frac{b}{2}\right)$ with gradient -2 : $y - \frac{b}{2} = -2(x - (b - 2))$, so

$$y = -2x + \frac{5b}{2} - 4.$$

7. Find b

Both equations represent l_2 , so $2b = \frac{5b}{2} - 4$. Hence $b = 8$.

Final answer

$$(a)(i) \ y = \frac{1}{2}x + 1. \quad (a)(ii) \ P = \left(b - 2, \frac{b}{2}\right).$$

$$(b) \ y = -2x + 2b.$$

$$(d) \ b = 8.$$